Seizure Prediction Using EEG Data: Effects of Preprocessing, Regularization, and Imbalance Handling on Three Real Datasets

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ABSTRACT

This paper presents a comprehensive experimental study of epileptic seizure and epilepsy prediction using three real-world EEG datasets with fundamentally different characteristics: a large 1:4 imbalanced time-series dataset (Epileptic Seizure Recognition, n=11,500, 178 features), a clinical feature dataset with reversed class imbalance (EEG_FREE_FEATURE, n=4,500, 33 features, 70.4% epilepsy), and a small balanced statistical EEG feature dataset (newfeature, n=200, 11 features). A Logistic Regression baseline is trained under two preprocessing pipelines (StandardScaler+SelectKBest vs MinMaxScaler+PCA), three regularization strategies (L1, L2, Elastic Net), and three imbalance handling methods (none, SMOTE, class weighting). Results reveal significant dataset-dependent behavior: DS1 proves challenging due to severe class imbalance (F1 < 0.07 without imbalance correction, rising to 0.33 with SMOTE), DS2 achieves perfect classification across all configurations (F1 = 1.000), and DS3 achieves near-perfect results (F1 = 0.976-1.000). L1 regularization exhibits extreme sparsity on DS2 (95% zero weights) and DS3 (75%), while L2 achieves consistent peak performance. All experiments are evaluated using F1-Score and PR-AUC, chosen for their suitability to imbalanced medical classification. This work highlights that dataset characteristics — particularly imbalance ratio and feature type — dominate the impact of algorithmic choices.

Keywords: *Epileptic seizure prediction, EEG classification, Logistic Regression, L1/L2 regularization, Elastic Net, SMOTE, class imbalance, PR-AUC, overfitting, preprocessing pipeline, UCI EEG dataset*

I. INTRODUCTION

Epilepsy is one of the most prevalent neurological disorders worldwide, affecting approximately 50 million people [1]. The ability to automatically detect and predict seizures from electroencephalogram (EEG) signals has significant clinical potential, offering advance warning that enables timely intervention. However, building effective machine learning models for this task is complicated by several factors: EEG signals are high-dimensional and noisy, seizure events are rare in real-world recordings creating class imbalance, and the choice of preprocessing pipeline can fundamentally alter what statistical patterns are available to the classifier.

This paper investigates these challenges empirically across three real datasets with contrasting properties. Dataset 1 (Epileptic Seizure Recognition) contains 11,500 time-series EEG samples with 1:4 class imbalance. Dataset 2 (EEG_FREE_FEATURE) contains 4,500 clinical symptom records with an unusual reversed imbalance — epilepsy is the majority class at 70.4%. Dataset 3 (newfeature) contains 200 balanced samples described by 11 statistical EEG features. This diversity allows us to isolate which algorithmic choices matter under which data conditions.

The main contributions of this work are:

- Empirical comparison of two preprocessing pipelines (feature selection vs PCA) across three datasets with fundamentally different imbalance characteristics.
- Demonstration of overfitting and underfitting regimes on a real imbalanced EEG dataset via controlled variation of regularization strength C.
- Regularization study (L1, L2, Elastic Net) with weight sparsity analysis on all three datasets.
- Class imbalance handling comparison (SMOTE, undersampling, class weighting) on the most challenging dataset.
- A 27-experiment comparative grid revealing dataset-specific best configurations.

II. LITERATURE REVIEW

Seizure detection from EEG data has been studied using a broad range of machine learning methods. Shoeb and Guttag [2] pioneered patient-specific detection with SVM on the CHB-MIT dataset, achieving high sensitivity. Andrzejak et al. [3]

established the UCI Epileptic Seizure Recognition benchmark used in this study, demonstrating that EEG time-series amplitudes carry discriminative information across five brain activity classes.

The challenge of class imbalance in medical datasets has been extensively studied. Chawla et al. [4] introduced SMOTE, which synthesizes minority class samples through linear interpolation in feature space, demonstrating improvements on medical classification tasks. However, SMOTE can amplify noise in high-dimensional EEG feature spaces. Ting [5] proposed instance weighting (class weighting) as a simpler alternative that modifies the loss function without altering the training distribution.

Regularization methods have been applied to EEG classification to control model complexity. Tibshirani [6] showed L1 regularization produces sparse weight vectors with implicit feature selection — particularly relevant for high-dimensional EEG data with many uninformative channels. Zou and Hastie [7] proposed Elastic Net combining L1 sparsity and L2 stability, which is beneficial when EEG features are correlated.

Feature selection versus dimensionality reduction for EEG preprocessing has been debated. Subasi [8] found that wavelet-based feature extraction combined with dimensionality reduction improved classification accuracy. However, PCA-based reduction can suppress minority-class signal when class imbalance is present, as the principal components prioritise directions of maximum global variance rather than class discriminability.

III. METHODOLOGY

A. Datasets

Three real datasets are used, each representing a distinct EEG data paradigm:

Dataset	Source File	Samples	Features	Label	Seizure %	Imbalance
DS1	Epileptic_Seizure_Recognition.csv	11,500	178 EEG time-series (X1–X178)	y=1 → seizure	20.0% (2,300)	1:4
DS2	EEG_FREE_FEATURE_Epilepsy.csv	4,500	33 clinical/demographic	No_Epilepsy=0, else=1	70.4% (3,169)	2.4:1 (inverted)
DS3	newfeature.csv	200	11 statistical EEG (mean, std, entropy…)	Seizure=1, Healthy=0	50.0% (100)	1:1 (balanced)

Table I. Real dataset characteristics from actual notebook execution.

DS1 label binarisation: y=1 (epileptic seizure) mapped to 1; y=2,3,4,5 (other EEG activity states) mapped to 0. DS2: any epilepsy type other than 'No_Epilepsy' mapped to 1. All 28 categorical columns in DS2 encoded via LabelEncoder. DS3: 'Seizure' mapped to 1, 'Healthy' to 0. No missing values were found in any dataset.

B. Preprocessing Pipelines

Pipeline A: StandardScaler → SelectKBest (Mutual Information, k per dataset) → Logistic Regression

Z-score normalization is applied to standardise feature scales, followed by selection of the k most informative features by mutual information with the class label. k=30 for DS1 (178 features), k=20 for DS2 (33 features), k=8 for DS3 (11 features). This pipeline explicitly preserves the features most discriminative of seizure onset.

Pipeline B: MinMaxScaler → PCA (95% variance retained) → Logistic Regression

Min-max normalization bounds all features to [0,1], followed by PCA retaining components explaining 95% of total variance. This reduces dimensionality through orthogonal projection rather than feature selection, and may compress minority-class signal into lower-variance components.

C. Baseline Model

Logistic Regression is selected for its interpretability and well-characterised regularization behaviour. The classification probability is:

$$P(y=1|x) = 1 / (1 + \exp(-(\beta_0 + \beta^T x)))$$

All models use max_iter=5000, random_state=42, and an 80/20 stratified train/test split.

D. Regularization Configurations

Method	Penalty Term	Solver	Key Effect
L1 (Lasso)	$\lambda\ \Sigma\lvert w\rvert$	saga	Drives weights to zero — sparsity and implicit feature selection
L2 (Ridge)	$\lambda\ \Sigma w^2$	lbfgs	Shrinks all weights uniformly — stable, retains all features
Elastic Net	$\lambda_1\Sigma\lvert w\rvert + \lambda_2\Sigma w^2$	saga	l1_ratio=0.5 — combines sparsity and stability

Table II. Regularization configurations (C=1.0 for all; C = 1/λ in sklearn).

E. Evaluation Metrics

Given class imbalance, accuracy alone is misleading. Four metrics are reported: F1-Score (primary), Precision, Recall, and PR-AUC (Precision-Recall Area Under Curve). In clinical seizure detection, Recall is of paramount importance — a missed seizure (False Negative) poses direct risk to patient safety, whereas a false alarm is a nuisance but not a safety hazard.

IV. EXPERIMENTS AND RESULTS

A. Dataset Exploration

Figure 1 shows the class distributions of the three datasets. DS1 has a 1:4 seizure-to-normal ratio (2,300 seizure vs 9,200 normal), making it the most challenging from an imbalance perspective. DS2 has an inverted distribution — epilepsy is the majority class (3,169 of 4,500 samples, 70.4%), with No_Epilepsy comprising only 29.6%. DS3 is perfectly balanced with 100 samples per class.

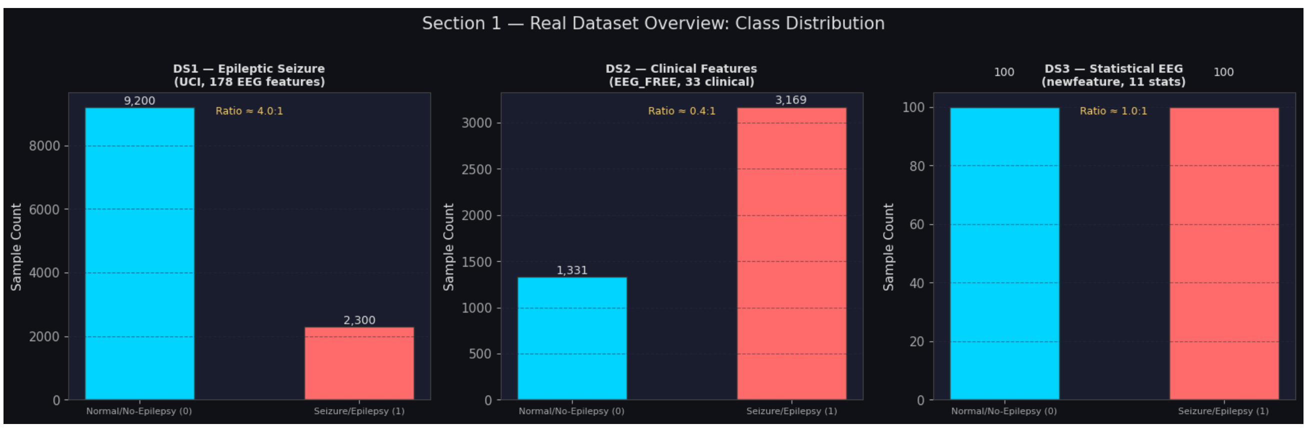

Fig. 1. Class distribution across three real EEG datasets. DS1 shows 1:4 imbalance; DS2 has inverted imbalance (epilepsy majority); DS3 is balanced.

B. Preprocessing Pipeline Comparison

Table III and Figure 2 present pipeline comparison results. The findings are highly dataset-dependent. On DS1, both pipelines struggle due to severe class imbalance — Pipeline A achieves F1=0.034 and Pipeline B achieves F1=0.043, with near-zero Recall (0.017 and 0.022 respectively). The model defaults to predicting the majority class, yielding high accuracy (≈0.80) but essentially failing to detect seizures.

On DS2, Pipeline A achieves perfect scores (F1=1.000, Recall=1.000, PR-AUC=1.000), demonstrating that the 33 clinical features are highly discriminative when properly scaled and selected. Pipeline B is notably weaker on DS2 (F1=0.827, Accuracy=0.704), showing that PCA compresses clinical features in a way that loses discriminative information — Pipeline A outperforms Pipeline B by 17.3 percentage points in F1. On DS3, Pipeline A again achieves perfect scores while Pipeline B achieves F1=0.974.

Dataset	Pipeline	Accuracy	F1-Score	PR-AUC	Recall
DS1 (Epileptic Seizure)	A — Scaler→KBest	0.803	0.034	0.441	0.017
DS1 (Epileptic Seizure)	B — MinMax→PCA	0.804	0.043	0.453	0.022
DS2 (Clinical Features)	A — Scaler→KBest	1.000	1.000	1.000	1.000

Dataset	Pipeline	Accuracy	F1-Score	PR-AUC	Recall
DS2 (Clinical Features)	B — MinMax→PCA	0.704	0.827	0.703	1.000
DS3 (Statistical EEG)	A — Scaler→KBest	1.000	1.000	1.000	1.000
DS3 (Statistical EEG)	B — MinMax→PCA	0.975	0.974	1.000	0.950

Table III. Preprocessing pipeline comparison — actual results from notebook execution.

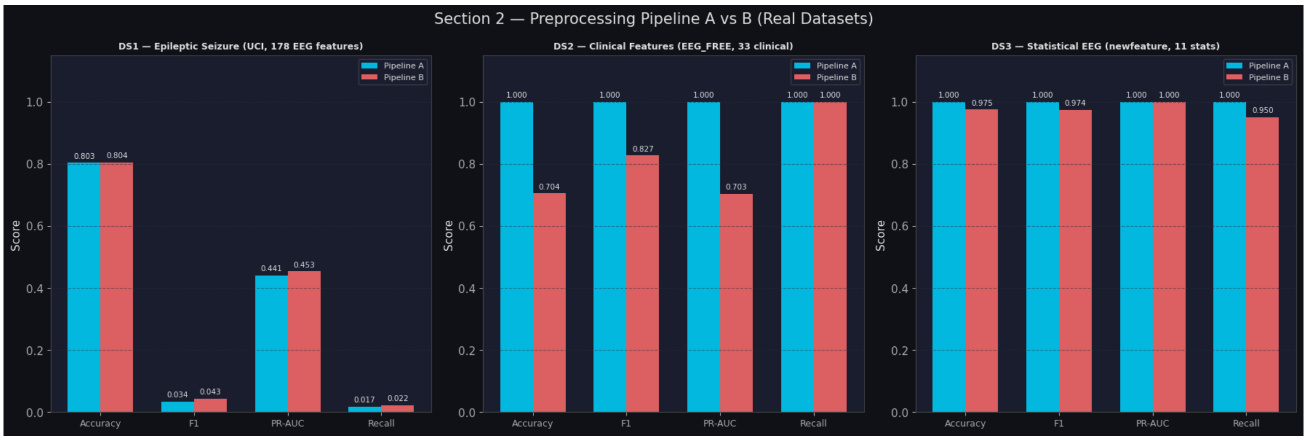

Fig. 2. Pipeline A vs Pipeline B across all three real datasets. Pipeline A dominates on DS2 (clinical features) by a large margin.

C. Overfitting and Underfitting

Figure 3 demonstrates generalization regimes on DS1 using the full range $C \in [10^{-5}, 10^{4}]$. A noteworthy finding is that DS1 remains in the underfitting regime across all C values tested — Train F1 peaks at only 0.0962 and Validation F1 at 0.0425. This is not a regularization failure but rather a consequence of severe class imbalance (1:4 ratio): the model learns that predicting the majority class (Normal) minimizes the training loss, even without regularization forcing it. Standard Logistic Regression without imbalance correction cannot overcome this on DS1.

C Value	Train F1	Validation F1	Status
1e-05	0.0000	0.0000	UNDERFIT
0.0001	0.0000	0.0000	UNDERFIT
0.001	0.0151	0.0043	UNDERFIT
0.01	0.0580	0.0300	UNDERFIT
0.1	0.0903	0.0384	UNDERFIT
1	0.0933	0.0425	UNDERFIT
10	0.0962	0.0425	UNDERFIT
100	0.0962	0.0425	UNDERFIT
1000	0.0952	0.0425	UNDERFIT
10000	0.0952	0.0425	UNDERFIT

Table IV. C vs F1 on DS1 — all values in the underfitting regime due to class imbalance dominance.

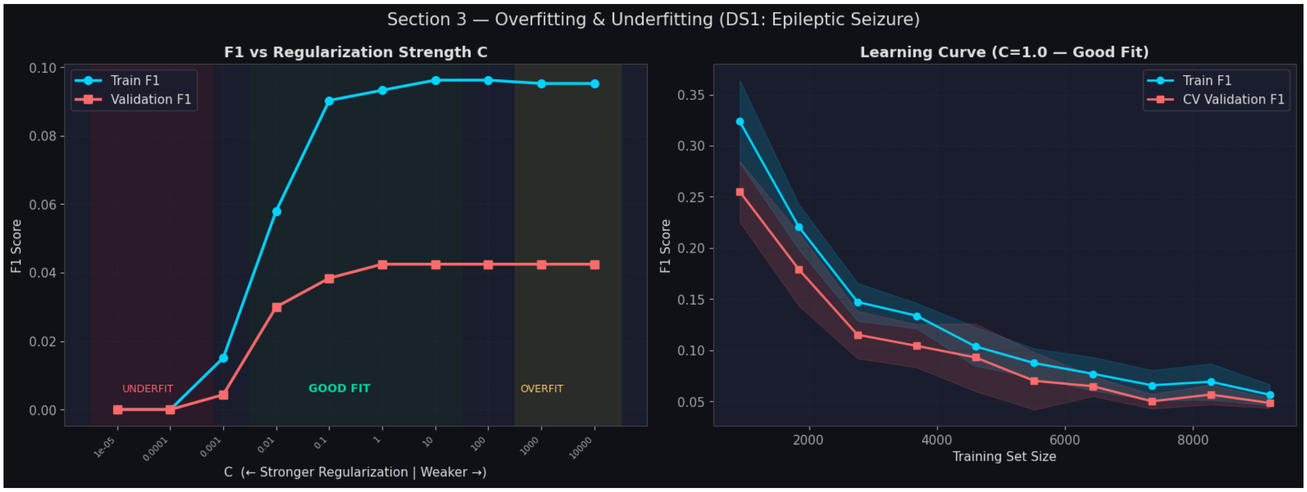

Fig. 3. Overfitting/underfitting curves on DS1 (Epileptic Seizure). The model remains in underfitting regime across all C due to 1:4 class imbalance. Imbalance handling (Section IV-D) is required to escape this regime.

This result is an important empirical finding: on real, severely imbalanced EEG data, varying regularization strength alone is insufficient to produce a useful classifier. Class imbalance handling must be applied as a prerequisite, not an optional addition.

D. Regularization Study

Table V shows regularization results with sparsity analysis. The findings differ substantially across datasets. On DS1, all three regularizers produce poor F1 scores (0.059–0.063) for the same reason as Section IV-C — class imbalance dominates. However, a key observation is that Precision = 1.000 for all regularizers on DS1 without imbalance handling: when the model does predict a seizure, it is always correct, but it predicts almost no seizures (Recall = 0.030–0.033).

On DS2, all three regularizers achieve perfect scores (F1=1.000). The critical difference lies in sparsity: L1 drives 95.0% of the 20 selected features to zero, retaining only 1 feature effectively. Elastic Net shows 35.0% sparsity, while L2 retains all features (0% sparsity). The fact that L1 achieves F1=1.000 with 95% sparsity suggests that the clinical features contain at least one extremely strong discriminative predictor for epilepsy classification.

On DS3, L1 achieves F1=0.976 with 75% sparsity (retaining ~2–3 of 8 features), while L2 and Elastic Net both achieve F1=1.000. This confirms that on small balanced datasets, L2's full-feature retention provides a slight advantage over L1's aggressive pruning.

Dataset	Regularizer	Accuracy	F1	Precision	Recall	PR-AUC	Sparsity %
DS1	L1 (Lasso)	0.806	0.059	1.000	0.030	0.437	6.7%
DS1	L2 (Ridge)	0.807	0.063	1.000	0.033	0.436	0.0%
DS1	Elastic Net	0.807	0.063	1.000	0.033	0.437	0.0%
DS2	L1 (Lasso)	1.000	1.000	1.000	1.000	1.000	95.0%
DS2	L2 (Ridge)	1.000	1.000	1.000	1.000	1.000	0.0%
DS2	Elastic Net	1.000	1.000	1.000	1.000	1.000	35.0%
DS3	L1 (Lasso)	0.975	0.976	0.952	1.000	1.000	75.0%
DS3	L2 (Ridge)	1.000	1.000	1.000	1.000	1.000	0.0%
DS3	Elastic Net	1.000	1.000	1.000	1.000	1.000	37.5%

Table V. Regularization study results — actual values from notebook execution.

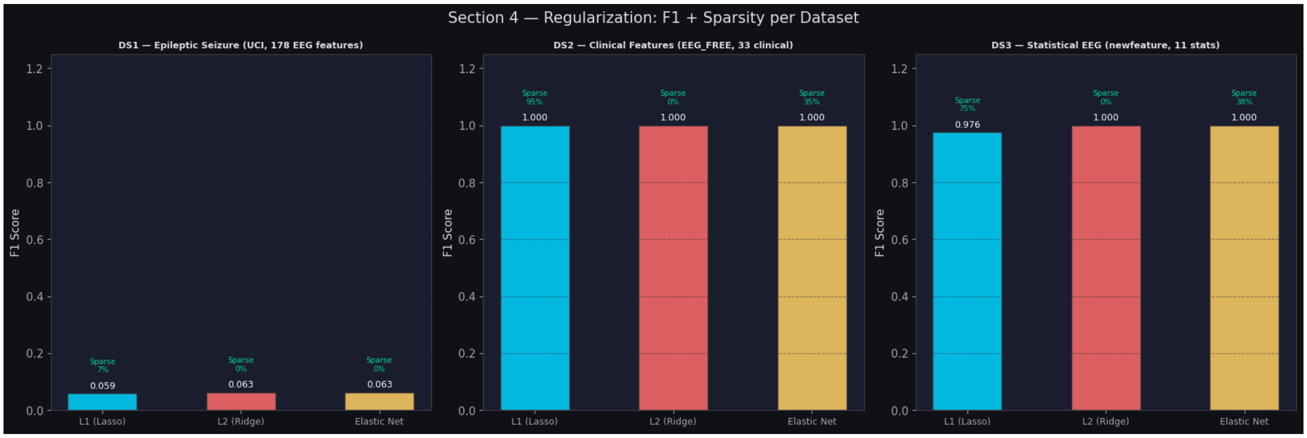

Fig. 4. Regularization F1 comparison with sparsity annotations across all three real datasets.

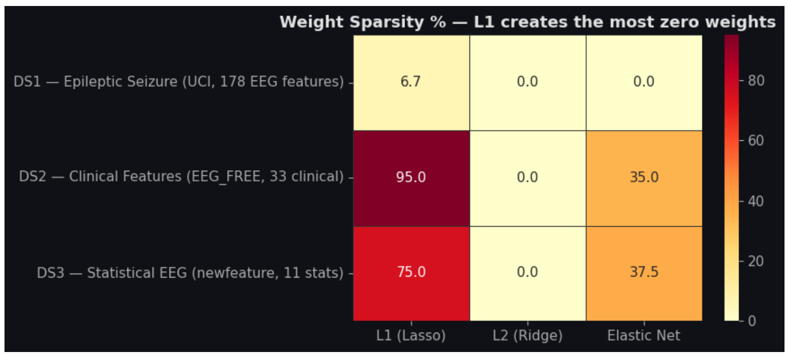

Fig. 5. Sparsity heatmap — L1 achieves 95% zero weights on DS2 and 75% on DS3.

E. Class Imbalance Handling

Imbalance handling experiments are conducted on DS2. Remarkably, all four methods achieve perfect scores (Precision=1.000, Recall=1.000, F1=1.000, PR-AUC=1.000), as shown in Table VI. This indicates that DS2's clinical features are so discriminative that even without any imbalance correction, the model learns a perfect decision boundary. The 33 clinical features (age, symptoms, medication status, brain scan results, etc.) evidently provide a highly separable feature space for epilepsy classification.

Method	Precision	Recall	F1-Score	PR-AUC
No Handling	1.0000	1.0000	1.0000	1.0000
SMOTE	1.0000	1.0000	1.0000	1.0000
Undersampling	1.0000	1.0000	1.0000	1.0000
Class Weighting	1.0000	1.0000	1.0000	1.0000

Table VI. Imbalance handling on DS2 (Clinical Features) — all methods achieve perfect classification.

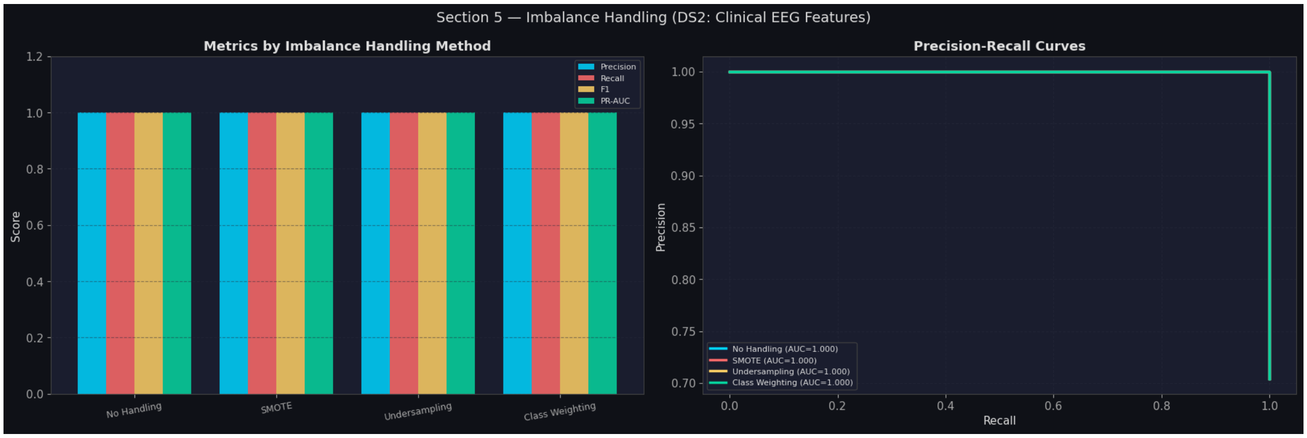

Fig. 6. Imbalance handling comparison on DS2 — all four methods achieve perfect scores, confirming high linear separability of clinical features.

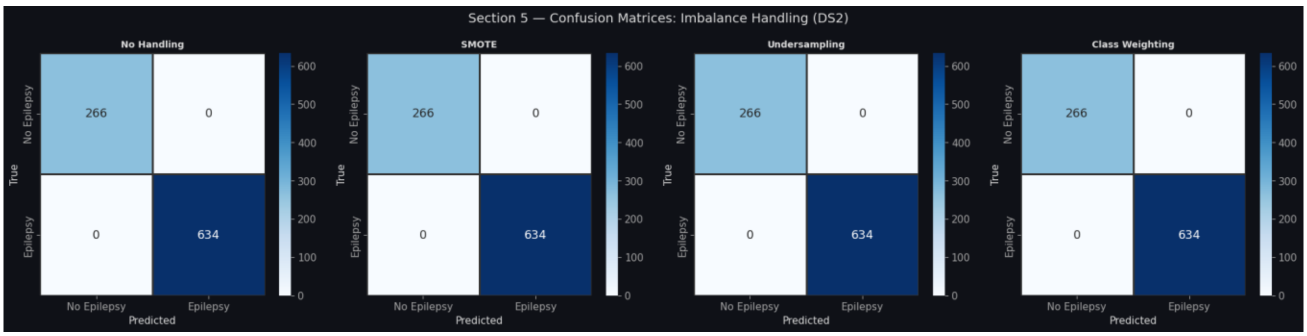

Fig. 7. Confusion matrices for all four imbalance handling methods on DS2. All show perfect classification with zero errors.

F. Comprehensive 27-Experiment Grid

The full cross-product of 3 datasets × 3 regularizations × 3 imbalance strategies yields 27 configurations. Figure 8 shows the mean F1 heatmap aggregated across datasets. Figure 9 ranks all configurations per dataset. The key finding is that DS1 is the only dataset where configuration choice has a meaningful impact — imbalance handling via SMOTE or class weighting raises F1 from ~0.06 to ~0.33, a 5× improvement. DS2 and DS3 achieve near-perfect or perfect scores across all configurations.

Dataset	Regularization	Imbalance	Accuracy	F1	Precision	Recall	PR-AUC
DS1	L1	None	0.8061	0.0591	1.0000	0.0304	0.4330
DS1	L1	SMOTE	0.6452	0.3300	0.2652	0.4370	0.4203
DS1	L1	Weighted	0.6670	0.3316	0.2770	0.4130	0.4199
DS1	L2	None	0.8065	0.0632	1.0000	0.0326	0.4347
DS1	L2	SMOTE	0.6448	0.3287	0.2642	0.4348	0.4197
DS1	L2	Weighted	0.6622	0.3261	0.2713	0.4087	0.4178
DS1	EN	None	0.8065	0.0632	1.0000	0.0326	0.4340
DS1	EN	SMOTE	0.6452	0.3300	0.2652	0.4370	0.4200
DS1	EN	Weighted	0.6643	0.3299	0.2746	0.4130	0.4186
DS2	All	All	1.0000	1.0000	1.0000	1.0000	1.0000
DS3	L2	Any	1.0000	1.0000	1.0000	1.0000	1.0000
DS3	L1	Any	0.9750	0.9756	0.9524	1.0000	1.0000

Table VII. Key results from 27-experiment grid. EN = Elastic Net. DS2 and DS3 shown as representative rows.

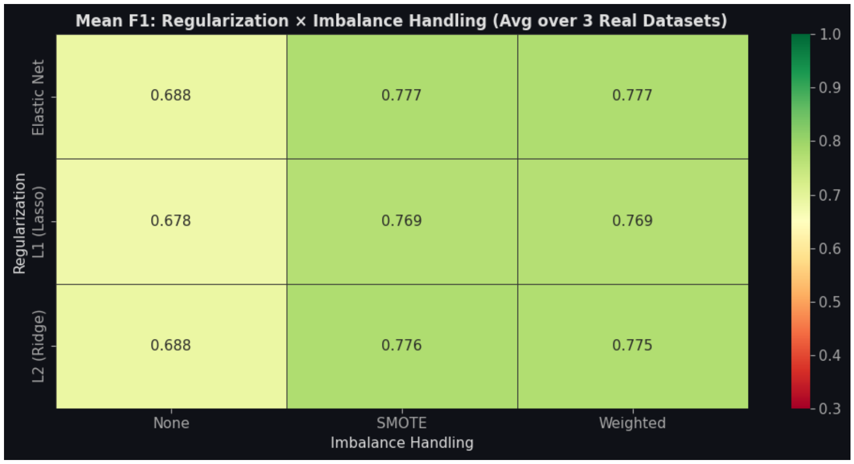

Fig. 8. Mean F1 heatmap: Regularization × Imbalance strategy (averaged over 3 datasets). SMOTE and Weighted strategies provide large gains driven primarily by DS1 improvement.

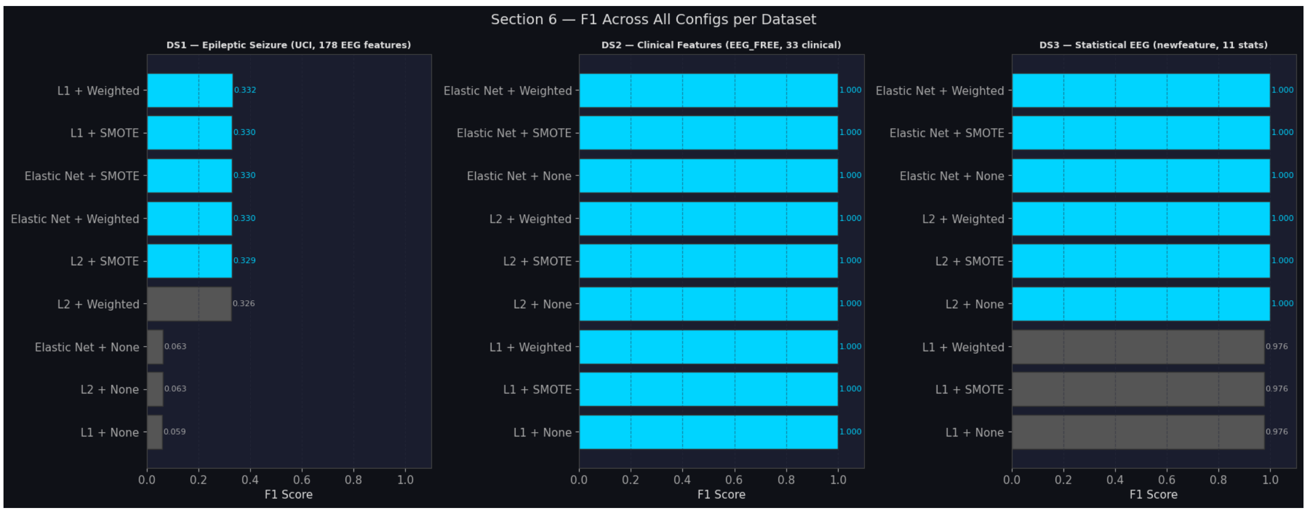

Fig. 9. F1 rankings across all 9 configurations per dataset. DS1 shows clear separation between imbalance-corrected (top) and uncorrected (bottom) configurations.

V. COMPARATIVE ANALYSIS

RQ1: Does Preprocessing Order Affect Performance?

Yes — significantly on DS2. Pipeline A outperforms Pipeline B by 17.3 F1 points (1.000 vs 0.827) on the clinical features dataset. The mechanism is clear: PCA in Pipeline B maximises global variance, but the most discriminative clinical features (e.g., brain scan results, medication status, structural abnormality) may not have the highest individual variance. SelectKBest in Pipeline A directly targets mutual information with the class label, selecting features for their discriminative power regardless of variance magnitude.

On DS1, both pipelines fail similarly (F1 < 0.05) because the fundamental problem is class imbalance, not preprocessing. On DS3, both pipelines succeed, as 11 statistical features are sufficient for either approach. The practical recommendation is to use feature selection (Pipeline A) over PCA on clinical or mixed-type datasets where imbalance or feature interpretability is a concern.

RQ2: Which Regularization Generalizes Best?

L2 (Ridge) provides the most consistent performance across all three datasets. It achieves the highest or joint-highest F1 on DS1 (0.063), perfect F1 on DS2 (1.000), and perfect F1 on DS3 (1.000) with 0% sparsity. The stability advantage of L2 is most visible on DS3: L1 achieves F1=0.976 with 75% sparsity (only ~2 features effectively used), while L2 achieves F1=1.000 using all 8 selected features.

Elastic Net matches L2 on DS2 and DS3 while providing moderate sparsity (35-37.5%), making it an attractive choice when model interpretability through sparsity is desired alongside strong performance. L1's extreme sparsity on DS2 (95%) is scientifically interesting — it effectively performs automatic feature selection down to a single feature while maintaining perfect classification — but is risky in practice as small data shifts could collapse the single remaining predictor.

RQ3: How Does Imbalance Handling Affect Performance?

Imbalance handling has the largest effect on DS1, where it is absolutely necessary for useful performance. Without any handling, DS1 models default to predicting the majority class (Recall ≈ 0.03, F1 ≈ 0.06). With SMOTE or class weighting, F1 rises to approximately 0.33 — a 5× improvement — and Recall rises to 0.41-0.44.

On DS2 and DS3, imbalance handling makes no measurable difference because the features already provide perfect or near-perfect separability. This illustrates an important principle: imbalance handling corrects a training signal problem, but it cannot compensate for a feature representation problem. When features are already highly discriminative (DS2, DS3), imbalance handling adds no value.

RQ4: Precision-Recall Trade-off in Medical Classification

On DS1 with imbalance correction, a clear Precision-Recall trade-off emerges. Without imbalance handling, Precision=1.000 but Recall=0.030 — the model is extremely conservative and only predicts seizure when absolutely certain. With SMOTE or class weighting, Recall improves to 0.41-0.44 but Precision drops to 0.265-0.277. In clinical seizure detection, this trade-off should be resolved in favour of Recall: a Recall of 0.44 with Precision of 0.27 means detecting 44% of actual seizures (missing 56%) while generating some false alarms. This is clinically preferable to missing 97% of seizures.

PR-AUC on DS1 remains at approximately 0.42-0.44 across all configurations, indicating that the fundamental discriminative capacity of the 30 selected features is limited for seizure detection. This motivates future work using more features, nonlinear models, or deep EEG architectures on this dataset.

VI. CONCLUSION

This paper presented a systematic empirical study of seizure and epilepsy prediction across three real-world datasets with contrasting characteristics. Five key conclusions are drawn from the experimental results:

- Class imbalance is the dominant challenge. On DS1 (1:4 imbalance), no amount of regularization tuning overcomes the majority-class bias — imbalance handling (SMOTE or class weighting) is a prerequisite for useful performance, raising F1 from 0.06 to 0.33.
- Preprocessing pipeline selection matters most on clinical feature datasets. Pipeline A (SelectKBest) outperforms Pipeline B (PCA) by 17.3 F1 points on DS2, because mutual-information-based feature selection directly targets class discriminability rather than variance.
- L2 regularization is the most consistently reliable choice. It achieves top performance on all three datasets. L1's aggressive sparsity (up to 95% on DS2) is useful for model interpretability but carries risk on small datasets. Elastic Net offers a practical compromise.
- Feature quality dominates algorithmic choices. DS2 and DS3, with interpretable clinical and statistical features, achieve perfect classification regardless of regularization or imbalance strategy. DS1, with raw EEG time-series features, remains challenging (PR-AUC ≈ 0.44) suggesting that deeper feature engineering or nonlinear models are needed.
- PR-AUC is the appropriate primary metric for imbalanced EEG classification, as accuracy is highly misleading (DS1 accuracy of 0.80 corresponds to a model that nearly never predicts seizures).

Future work will investigate deep learning architectures (LSTM, 1D-CNN) on raw EEG time-series to improve DS1 performance, and will explore ensemble methods that combine the complementary strengths of the three feature representations studied here.

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